PROJECT #4 – INTERPRET

Client Prompt:

The National Risk Index (NRI) is a measurement conducted by the Department of Homeland Security for the Federal Emergency Management Agency (FEMA) to determine the relative risk of natural disaster events in the United States. Risk is classically defined as the product of probability of an event times the impact of the event should the event occur. The National Risk Index provides relative Risk Index percentiles and ratings based on data for Expected Annual Loss due to natural hazards, Social Vulnerability, and Community Resilience. Separate percentiles and ratings are also provided for each component: Expected Annual Loss, Social Vulnerability, and Community Resilience. For the Risk Index and Expected Annual Loss, percentiles and ratings can be viewed as a composite score for all hazards or individually for each of the 18 hazard types (FEMA National Risk Index, 2025). A community's score is represented by its percentile ranking among all other communities at the same level for Risk, Expected Annual Loss, Social Vulnerability and Community Resilience. For example, if a given Census tract's Risk Index percentile for a hazard type is 84.32 then its Risk Index value is greater than 84.32% of all US Census tracts. For Risk, Expected Annual Loss, Social Vulnerability, and Community Resilience, there is a qualitative rating that describes the community in comparison to all other communities at the same level, ranging from "Very Low" to "Very High." (FEMA National Risk Index, 2025). For more information on how risk is determined, please visit https://hazards.fema.gov/nri/understanding-scores-ratings

You have been hired by Risk Averse, LLC, an independent risk analysis consulting research group, to review where bias may exist in the NRI scoring methodology by performing what is known as a sensitivity analysis. A sensitivity analysis varies the input to a model or method and determines how the outputs change as a result. Risk Averse, LLC, wants to compare the current risk levels determined using the NRI method and data to an independent consultant's definition of risk. Risk Averse, LLC, has asked you to perform research to create an alternative risk scoring method that can use the NRI dataset as an input. You will compare the NRI to your risk index and determine if there is "categorical bias" in how risk is determined by both methods.

Using both NRI's original definitions for risk and your new proposed definitions, Risk Averse, LLC has asked you and your team to conduct an analysis of the risk of natural disasters in two states of your choice. This will include the following elements:

1. Downloading, cleaning and organizing two state's data at the Census Tract level from the NRI website (https://hazards.fema.gov/nri/data-resources)
2. Downloading and using the metadata files from NRI to clarify multiple variables in the NRI dataset
3. Downloading and using the SVI dataset from ASTDR: https://www.atsdr.cdc.gov/place-health/php/svi/svi-data-documentation-download.html
4. Joining multiple datasets together from the NRI database to aid Risk Averse, LLC in future endeavors.
5. Handling missing or NA values in the dataset
6. Generating at least two summary tables or figures showing risk scoring by both the NRI definitions and your proposed definition
7. Generate at least one map using GeoPandas that communicates risk information to the client.

You and your team will need to download and clean the data for each of your two identified states at the Census Tract level and the metadata from the NRI website. You will need to do some additional research outside of the materials provided to generate your proposed risk definition. You will need to perform some basic research to understand how risk is defined, how the risk index relates to outcomes for communities; remember, the risk index is used to help identify priority communities that may need management strategies due to the potential for natural disasters in their locality.

Deliverables:

1. Raw and cleaned data files used in the Python code file
2. A Python code file that documents the code used to generate your solution
3. An annotated code document that explains each line of code in detail
4. A well-written scope of work related to the client prompt that defines the tasks undertaken by your group to complete the project. The scope of work must contain the following items:
    a. Identifies the two states you will be examining in your risk analysis study
    b. Identifies at least two external resources that will be useful in your analysis
    c. Reports the current NRI definitions of risk
    d. Explains how the current NRI is calculated using the NRI website links provided.
5. A Gantt chart of the activities completed as part of the project
6. A timesheet detailing hours spent on each activity and by whom within the group
7. A GitHub Repository link containing all code and project data files.
8. A written document summarizing the findings in the code that can be handed to your PE for later use
    a. Your report must include the following elements:
        i. A clear methodology for how you defined risk in your data analysis so that your results are reproducible
        ii. A written comparison of your risk definitions compared to the NRI definitions . Your comparison should include consideration of the ethical implications of risk definitions, examining factors such as (but not limited to) bias in risk definitions between different states, implications for future funding for Risk Averse, LLC and data collection activities and how states and communities may interpret these results and make decisions and policies.
        iii. A written comparison of the results found using the NRI risk definitions and your proposed risk definition, specifically examining what implications these differences will have on the people impacted by natural disasters and how this relates to being able to give communities FEMA funding in the future to mitigate disasters.

All files must be submitted to Canvas by Tuesday April 21st, 2026 at 5:00pm CT.

Skills Students will learn:

- Define types of bias in code
- Identify sources of bias in code
- Deconstruct Python code
- Recognize bias in Python code and understand how data analysis decision making can introduce bias into code
- Justify code to a client
- Verify code meets client needs
- Utilizing GeoPandas for mapping within Python

References

FEMA National Risk Index. (2025). Understanding Scores and Ratings. Retrieved from: https://hazards.fema.gov/nri/understanding-scores-ratings